

Evidence-driven dimensionality reduction

pbmc3k

Reporting mode: deterministic. This report is rendered from saved evidence by deterministic templates; no LLM API wrote this report.

Dataset and declared scope

Source: Scanpy datasets.pbmc3k and pbmc3k_processed; 10x PBMC3k. The inspected source has 2700 observations and 32738 numerical features. Sparsity: 0.9741. The pilot contains 1000 observations; the final analysis contains 2638 of 2638 available post-QC observations.

Class counts in the source: {'B cells': 342, 'CD14+ Monocytes': 480, 'CD4 T cells': 1144, 'CD8 T cells': 316, 'Dendritic cells': 37, 'FCGR3A+ Monocytes': 150, 'Megakaryocytes': 15, 'NK cells': 154, 'unannotated': 62}. Labels control stratified cohort composition and secondary diagnostic plots only. They do not enter preprocessing fits, embeddings, or primary selection.

Preprocessing and rationale

cell_qc: Remove low-information cells and extreme gene/mitochondrial counts. Parameters: min_genes=200; max_genes=2500; max_mito_fraction=0.05; retained=2638; . Fit scope: per-cell fixed thresholds.

sample_cohort: Common representative cohort; labels used only to preserve class proportions. Parameters: size=2638; seed=774; stratified=True; . Fit scope: cohort design.

filter_genes: Remove genes observed in too few cohort cells. Parameters: min_cells=3; retained=13656; . Fit scope: analysis cohort only.

normalize_log1p: Adjust library size and compress count range. Parameters: target_sum=10000.0; . Fit scope: analysis cohort only.

select_highly_variable_genes: Focus on variable expression features in sparse high-dimensional count data. Parameters: n_genes=2000; flavor=seurat; . Fit scope: analysis cohort only.

scale_clip: Balance gene scales and limit outliers. Parameters: max_value=10; . Fit scope: analysis cohort only.

common_pca_representation: Common denoising and computational input for all ten methods; final PCA is two-dimensional. Parameters: n_components=50; . Fit scope: analysis cohort only.

Comparable pilot design

Every method was attempted on identical saved input ([1000, 50]). MDS, GPLVM, t-SNE, and UMAP use repeated seeds; other methods run once. The metric reference is the same common PCA representation. Trustworthiness and mean k-nearest-neighbor recall use Euclidean distance and k=15; stability is mean pairwise seed neighborhood overlap. Quality = 0.45 trustworthiness + 0.45 neighbor recall + 0.10 stability. Deterministic stability is defined as one. Silhouette is secondary, excludes unannotated cells, and uses at most 2,000 observations.

Method	Status	Trust	Recall	Stability	Quality	Seconds
pca	success	0.768	0.090	1.000	0.486	1.810
kernel_pca	success	0.716	0.069	1.000	0.453	1.889
mds	success	0.765	0.097	0.271	0.415	8.063
isomap	success	0.787	0.105	1.000	0.501	2.028
lle	success	0.730	0.077	1.000	0.463	1.801
laplacian	success	0.769	0.111	1.000	0.496	1.743
diffusion_maps	success	0.720	0.071	1.000	0.456	2.817
gplvm	success	0.787	0.106	0.164	0.418	16.494
tsne	success	0.845	0.255	1.000	0.595	7.641
umap	success	0.810	0.193	0.321	0.483	24.098

Prior suitability, empirical decisions, and failures

Best pilot quality 0.594887; eligibility threshold 0.535398. Retained methods: pca, tsne, isomap, laplacian. Low-confidence fallback: False.

pca — retain. Prior: Linear variance baseline. Evidence and decision: Quality=0.486042; threshold=0.535398; runtime=1.810s; Pareto=False. Retained linear baseline.

kernel_pca — exclude. Prior: Nonlinear kernel structure; quadratic memory. Evidence and decision: Quality=0.453135; threshold=0.535398; runtime=1.889s; Pareto=False. Below quality/Pareto eligibility or outside configured candidate capacity.

mds — exclude. Prior: Global pairwise geometry; iterative quadratic distance work. Evidence and decision: Quality=0.414968; threshold=0.535398; runtime=8.063s; Pareto=False. Below quality/Pareto eligibility or outside configured candidate capacity.

isomap — retain. Prior: Geodesic geometry conditional on a connected neighborhood graph. Evidence and decision: Quality=0.501324; threshold=0.535398; runtime=2.028s; Pareto=True. Retained by quality/Pareto rank and capacity.

lle — exclude. Prior: Local affine reconstruction conditional on well-conditioned neighborhoods. Evidence and decision: Quality=0.462876; threshold=0.535398; runtime=1.801s; Pareto=False. Below quality/Pareto eligibility or outside configured candidate capacity.

laplacian — retain. Prior: Local graph geometry conditional on connectivity. Evidence and decision: Quality=0.496076; threshold=0.535398; runtime=1.743s; Pareto=True. Retained by quality/Pareto rank and capacity.

diffusion_maps — exclude. Prior: Density-corrected diffusion geometry. Evidence and decision: Quality=0.455610; threshold=0.535398; runtime=2.817s; Pareto=False. Below quality/Pareto eligibility or outside configured candidate capacity.

gplvm — exclude. Prior: Probabilistic nonlinear latent-variable model under a fixed optimization budget. Evidence and decision: Quality=0.418321; threshold=0.535398; runtime=16.494s; Pareto=False. Below quality/Pareto eligibility or outside configured candidate capacity.

tsne — retain. Prior: Stochastic local-neighborhood visualization. Evidence and decision: Quality=0.594887; threshold=0.535398; runtime=7.641s; Pareto=True. Retained by quality/Pareto rank and capacity.

umap — exclude. Prior: Stochastic neighborhood-graph embedding. Evidence and decision: Quality=0.483419; threshold=0.535398; runtime=24.098s; Pareto=False. Below quality/Pareto eligibility or outside configured candidate capacity.

Shortlist refinement and final results

Only retained methods were tuned on the pilot cohort. The selection metrics are reused, including repeated-seed stability; ties at six decimal places favor lower runtime and then grid order. These are in-cohort descriptive metrics, not held-out generalization estimates.

Method	Grid	Status	Quality	Seconds
pca	0	success	0.486	1.53
tsne	0	success	0.595	7.53
tsne	1	success	0.586	6.13
tsne	2	success	0.588	8.63
isomap	0	success	0.501	1.99
isomap	1	success	0.504	1.78
isomap	2	success	0.501	1.94
laplacian	0	success	0.496	1.74
laplacian	1	success	0.502	1.62
laplacian	2	success	0.492	1.70

Final cohort planning: {'configured_cap': 3000, 'declared_final_size': 2638, 'pilot_samples': 1000, 'estimates': {'pca': {'observed_pilot_seconds': 1.8095547519624233, 'conservative_sample_limit': 7923}, 'isomap': {'observed_pilot_seconds': 2.0282948869280517, 'conservative_sample_limit': 7627}, 'laplacian': {'observed_pilot_seconds': 1.7434960040263832, 'conservative_sample_limit': 8021}, 'tsne': {'observed_pilot_seconds': 7.64052322646603, 'conservative_sample_limit': 4901}}, 'rule': '2x pilot runtime * (N_final/N_pilot)^3 <= per-run timeout; cap; conservative planning bound, not measured scaling. PBMC normally full post-QC.'}.

Method	Trustworthiness	Neighbor recall	Silhouette (secondary)	Seconds
pca	0.825	0.051	0.404	1.52
tsne	0.897	0.208	0.419	16.30
isomap	0.822	0.051	0.329	2.92
laplacian	0.838	0.077	0.390	2.14

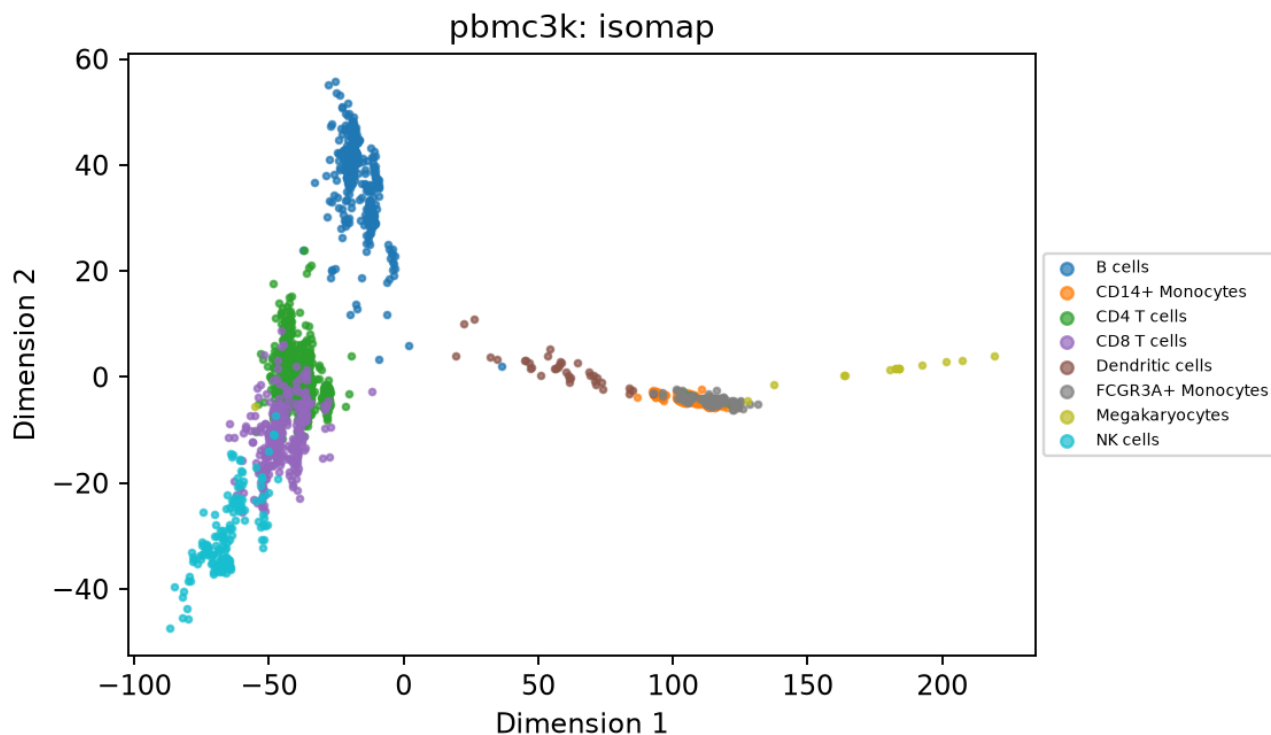
Interpretation and limitations

On this final cohort, tsne has the highest observed trustworthiness (0.897), while tsne has the highest neighbor recall (0.208). The first two PCA components explain 3.8% of variance in the preprocessed feature space. These diagnostics describe different aspects of the embedding and need not identify the same method as best.

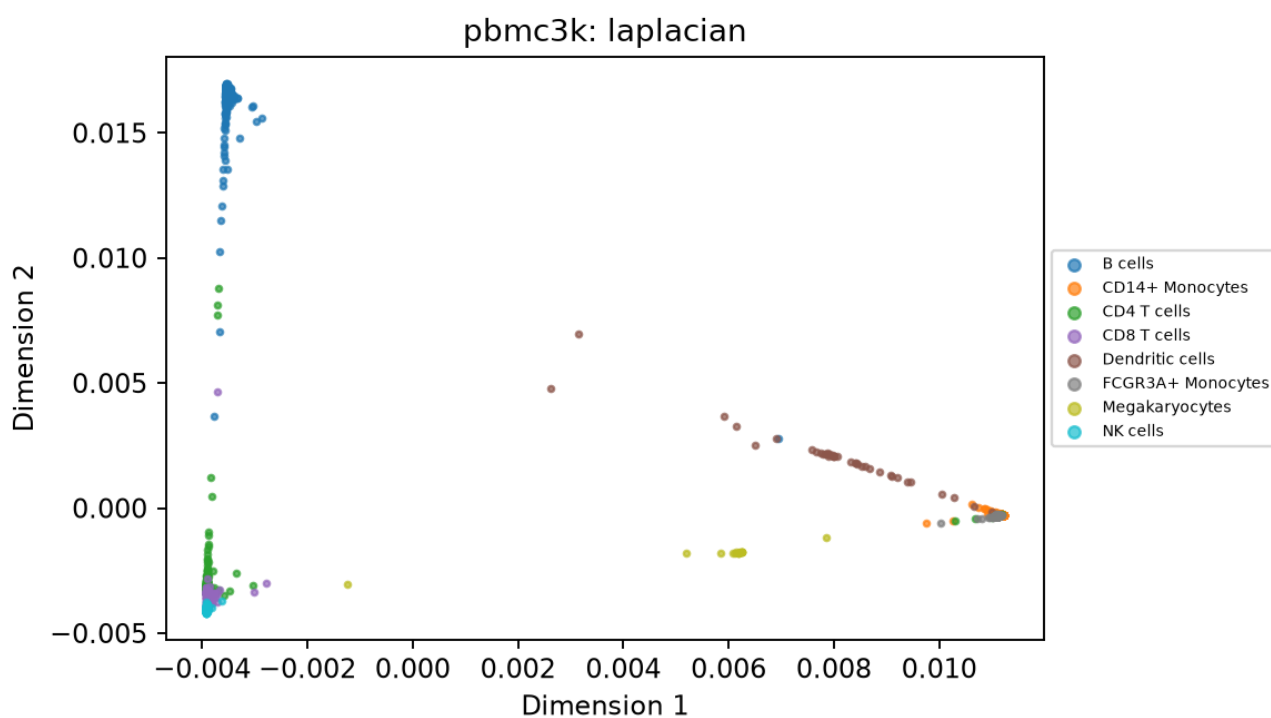
The shortlist reflects measured preservation of local geometry in the declared input representation under the specified compute budget. It does not establish a universally superior method. Changes in preprocessing, cohort composition, neighborhood scale, or objective weights can change the ranking. Linear PCA remains an interpretable baseline. Disconnected graphs and numerical warnings are retained in the evidence and affect eligibility.

GPLVM uses actual variational optimization with latent uncertainty and inducing GP variables. Its fixed budget does not prove convergence; loss history and latent displacement are saved. Successful-run runtime includes method imports, input loading and fitting, but excludes subprocess startup and common imports. Cold caches and hardware can affect timing. A conservative cubic bound informs PathMNIST cohort planning but is not a fitted complexity model. Biological and histological labels are diagnostics, not validation of causal or biological mechanisms. PBMC annotations derive from a prior processed clustering and are not independent ground truth.

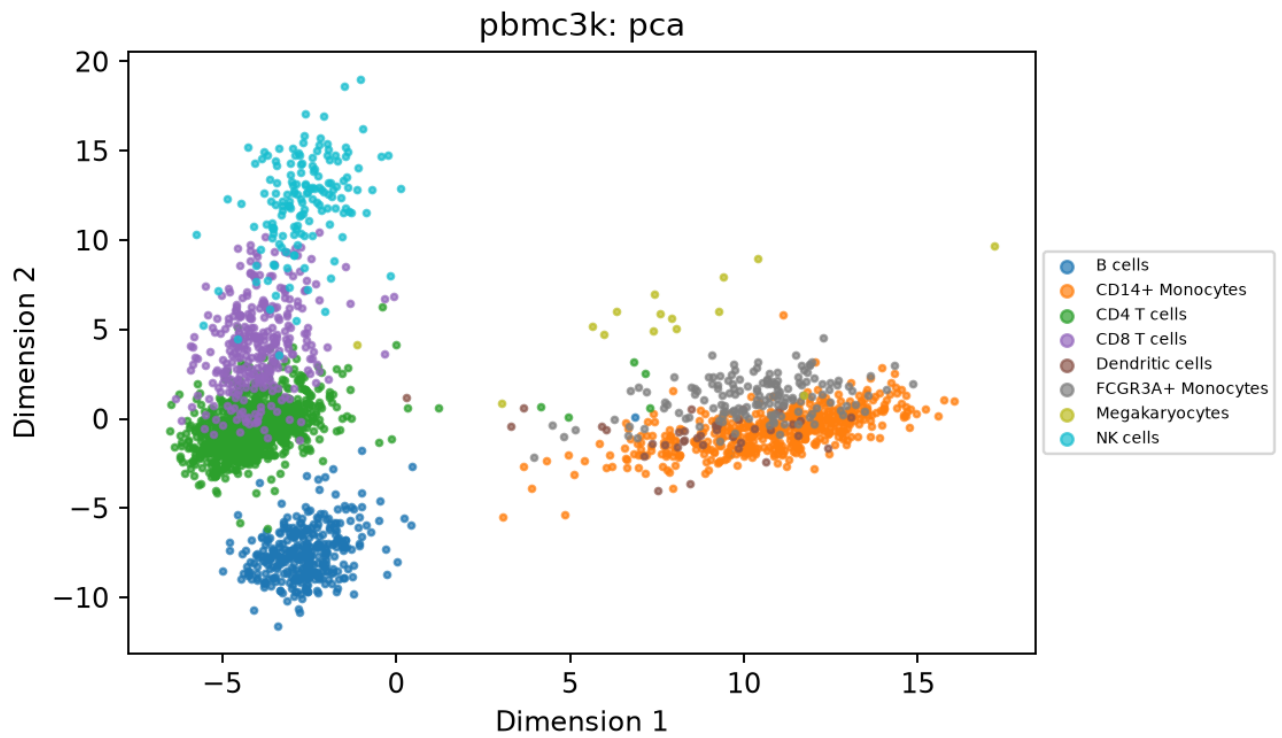
Figures



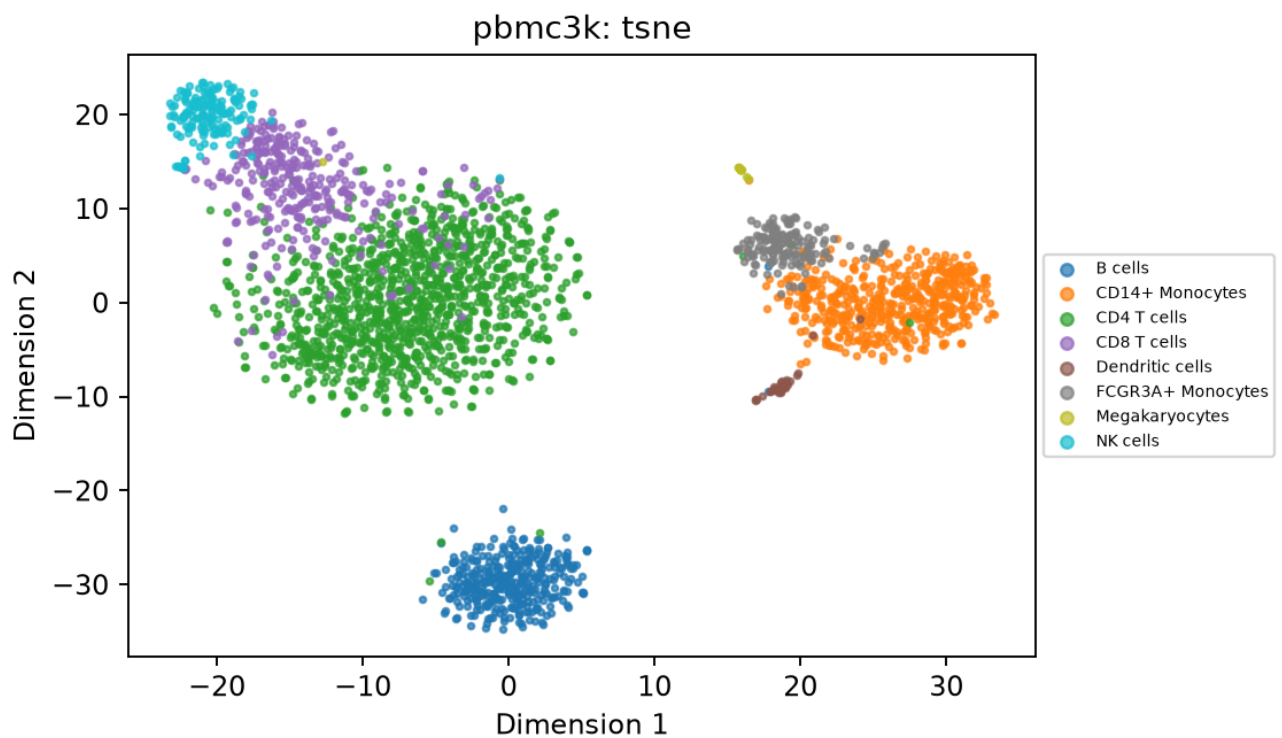
isomap on the declared final cohort. Colors are secondary labels; visual separation is not evidence of biological truth or generalization.



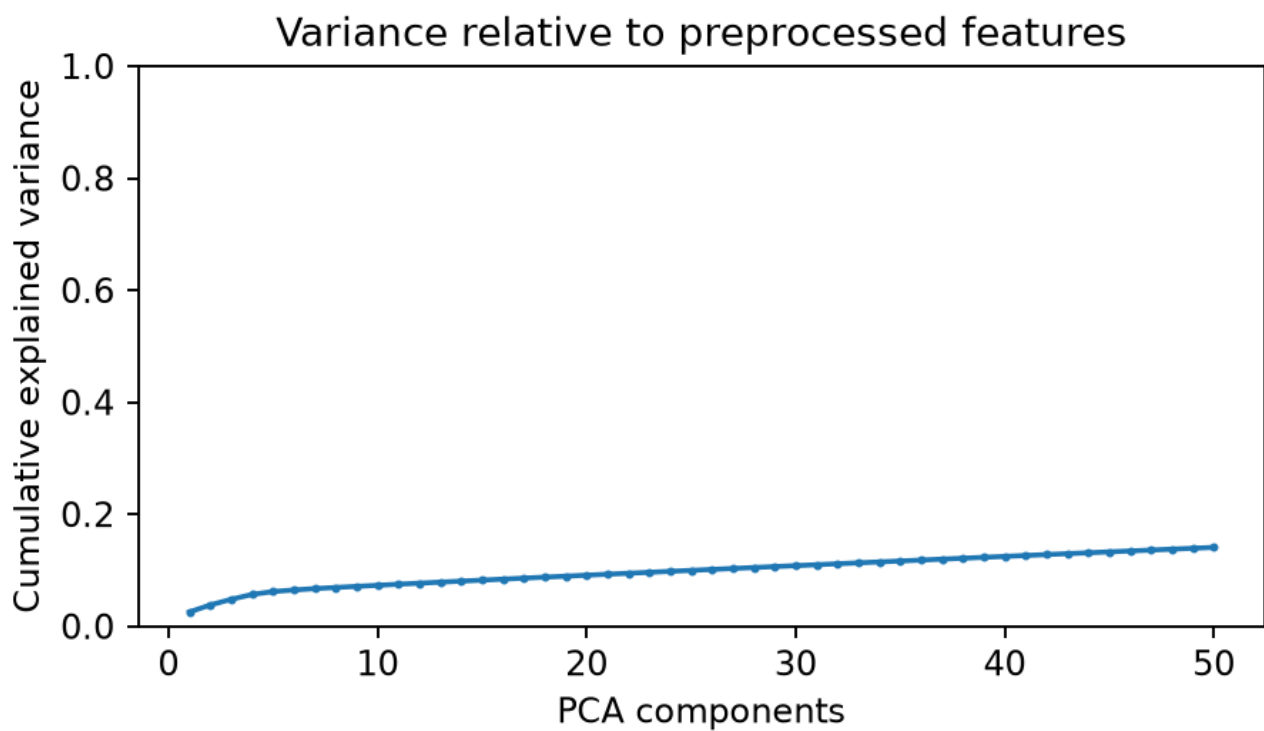
laplacian on the declared final cohort. Colors are secondary labels; visual separation is not evidence of biological truth or generalization.



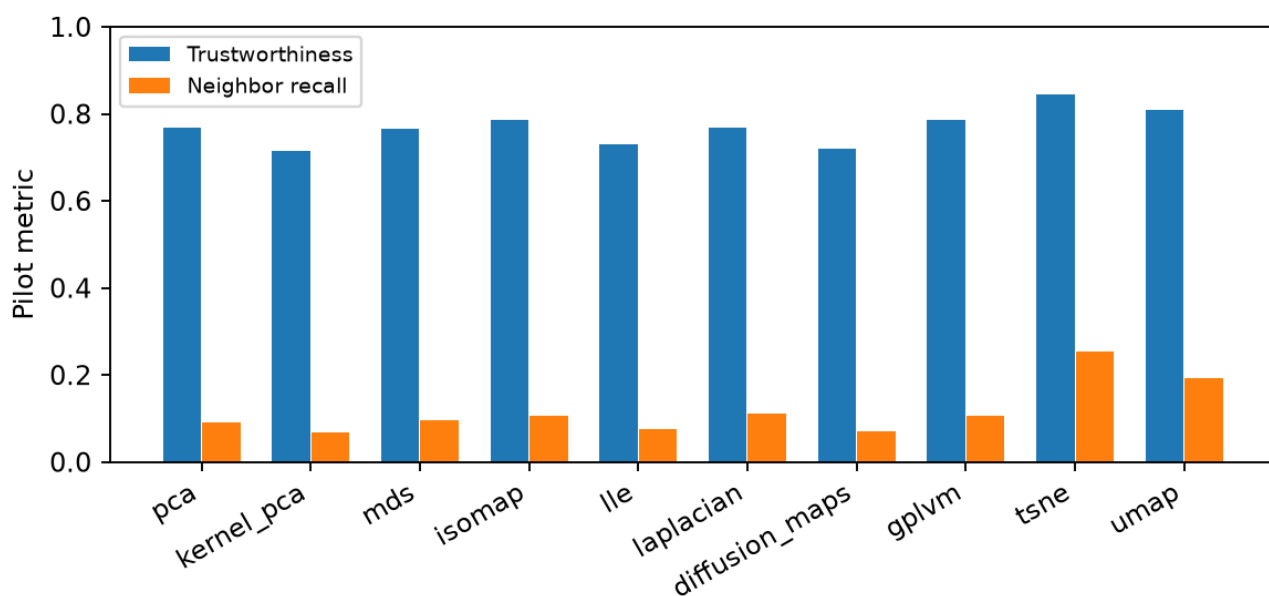
pca on the declared final cohort. Colors are secondary labels; visual separation is not evidence of biological truth or generalization.



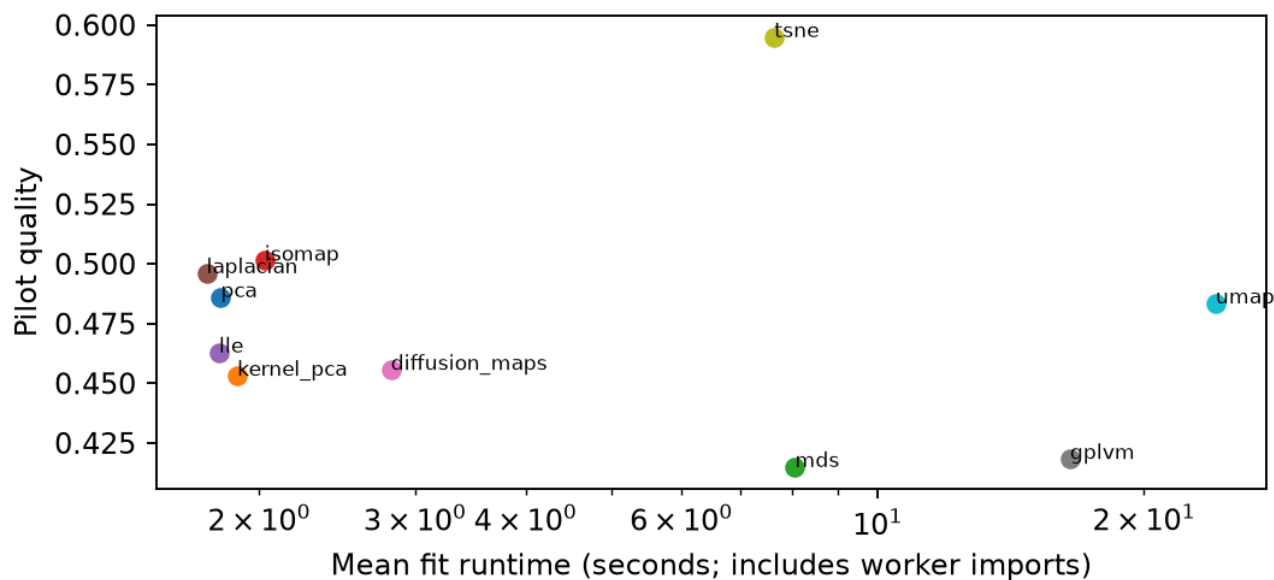
tsne on the declared final cohort. Colors are secondary labels; visual separation is not evidence of biological truth or generalization.



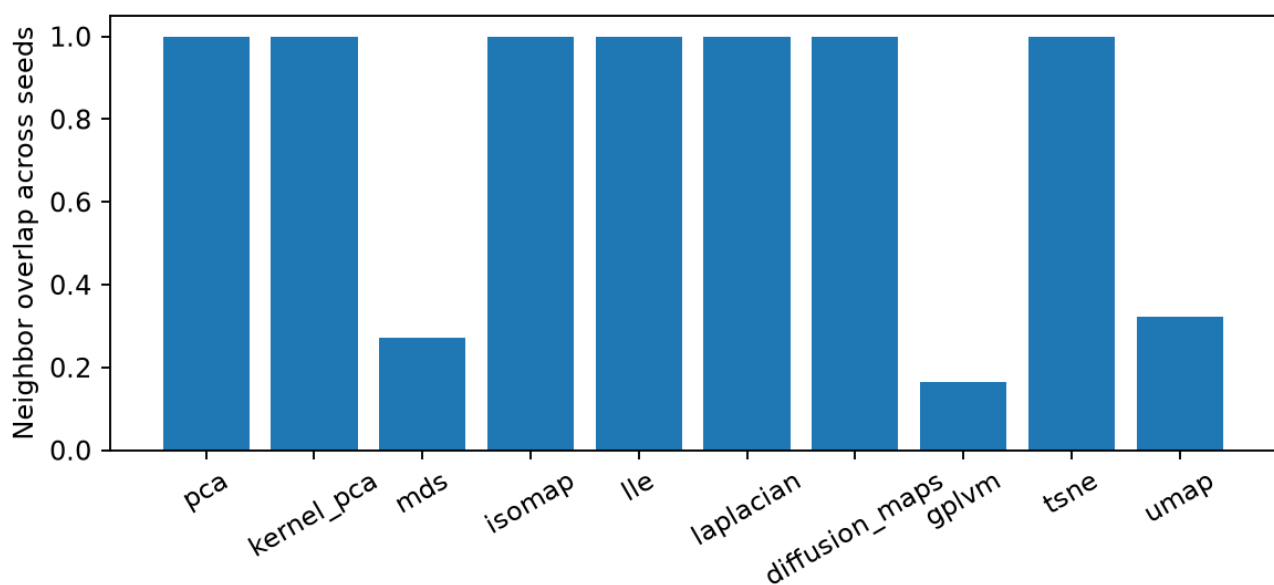
Variance retained by the common PCA input representation; first two components correspond to the displayed linear PCA baseline.



Metrics averaged across successful seeds on the same pilot cohort and reference.



Observed runtime against declared label-free quality; higher quality and lower runtime are favorable. Warnings also affect eligibility.



Mean pairwise seed neighborhood overlap; deterministic methods are defined as 1, not empirically replicated.

Method	Decision	Pilot status
pca	retain	success
kernel_pca	exclude	success
mds	exclude	success
isomap	retain	success
lle	exclude	success
laplacian	retain	success
diffusion_maps	exclude	success
gplvm	exclude	success
tsne	retain	success
umap	exclude	success

Candidate decisions; numerical reasons are recorded in decisions.json.

Reproducibility and sources

Configuration SHA256: 4f92f3ab707d97ef4db494f2c9b6184a7a25f22fbf55a3db6ac946f23bf3e46c

Code SHA256: 096a3ae17bbbc370843872200cad522f81a94de23f3b63d0ecc8db1c5a45c4aa

Git commit: None

Job: 2133669; host: c0405.ll.unc.edu; timestamp: 2026-09-23T00:24:26.046358+00:00.

Software: {'numpy': '2.4.6', 'pandas': '2.3.3', 'scipy': '1.17.1', 'scikit-learn': '1.9.1', 'anndata': '0.12.19', 'scanpy': '1.11.5', 'umap-learn': '0.5.12', 'pydiffmap': '0.2.0.1', 'medmnist': '3.0.2', 'torch': '2.14.0+cpu', 'gpytorch': '1.15.2', 'matplotlib': '3.11.2', 'weasyprint': '70.0', 'pypdf': '6.19.0', 'openai': '3.18.0'}

Source provenance: {'source': 'Scanpy datasets.pbmc3k and pbmc3k_processed; 10x PBMC3k', 'url': 'https://scanpy.readthedocs.io/en/latest/api/generated/scanpy.datasets.pbmc3k.html', 'access_date': '2026-09-23', 'files': {'pbmc3k_processed.h5ad': '0db367b991dd95809732b218539ede489bea99113807f62ebd7ccc970025fe38', 'pbmc3k_raw.h5ad': '89a96f1beaa2dd83a687666d3f19a4513ac27a2a2d12581fcd77afed7ea653a1'}, 'versions': {'numpy': '2.4.6', 'pandas': '2.3.3', 'scipy': '1.17.1', 'scikit-learn': '1.9.1', 'anndata': '0.12.19', 'scanpy': '1.11.5', 'umap-learn': '0.5.12', 'pydiffmap': '0.2.0.1', 'medmnist': '3.0.2', 'torch': '2.14.0+cpu', 'gpytorch': '1.15.2', 'matplotlib': '3.11.2', 'weasyprint': '70.0', 'pypdf': '6.19.0', 'openai': '3.18.0'}, 'annotation': 'processed louvain annotation aligned by exact barcode', 'unannotated': 62}

References: Scanpy PBMC3k dataset documentation; 10x Genomics PBMC3k. Yang et al., MedMNIST v2, Scientific Data (2023), doi:10.1038/s41597-022-01721-8. Titsias and Lawrence, Bayesian Gaussian Process Latent Variable Model, AISTATS (2010). GPyTorch GPLVM SVI tutorial. Full method and source links are in docs/METHOD_NOTES.md and docs/DATA_PROVENANCE.md.